



A natural disaster and intimate partner violence: Evidence over time

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ABSTRACT

Natural disasters affect about 200 million people annually. Heightened intimate partner violence (IPV) is a gendered impact of these disruptive events. This study examines prevalence and correlates of IPV in four Indian states—TamilNadu, Kerala, AndhraPradesh, and Karnataka—before and after the Indian Ocean tsunami of 2004. Drawing on three waves of National Family Health Surveys of India—six years before, immediately after, and a decade after disaster, this paper evaluates if TamilNadu and Kerala (severely affected) exhibited higher prevalence of IPV than AndhraPradesh (moderately affected) and Karnataka (not directly affected). Logistic regression analyses determine association between IPV, state of residence (proxy for experience of disaster), and other covariates. To test hypotheses guided by vulnerability theory, IPV was regressed on socio-economic and demographic predictors for states across waves. IPV increased by 48% between 2005 and 2015. Increase in physical (61%) and sexual (232%) violence was highest in TamilNadu; emotional violence increased by 122% in Karnataka. State of residence was associated with IPV in the aftermath of disaster. In 2005, compared to Karnataka, odds of IPV were 98% higher in TamilNadu and 41% higher in Kerala. A decade after, odds were two times higher in TamilNadu than in Karnataka. Belonging to disadvantaged groups predicted higher odds of IPV in the year after disaster. Higher socio-economic status predicted lower odds of IPV, except in Kerala. Data point to ways in which socio-economic and demographic vulnerabilities factor into risk of IPV after disaster. Demographic factors of religion and caste appear to lose significance over time, but socio-economic factors continue to matter. Disaster response strategies seldom work without tackling long-standing inequities. Appropriate support systems for women and minorities in non-disaster situations are critical to ensure their conditions are not exacerbated.

1. Introduction

Anthropogenic climate change is associated with increased intensity and frequency of natural disasters (IPCC, 2018; N.A.S., 2016). These disruptive events affect population groups differentially; higher mortality and violence among women and children, loss of social networks and social isolation among older adults, breakdown in systems of support leading to losses in education and care-mechanisms for children, are cases in point (Frankenberg et al., 2013; Harville et al., 2011; Khunwshit and McEntire, 2012; Masson et al., 2016).

Intimate partner violence (IPV) is a crucial public health issue across the world (Ackerson and Subramanian, 2008; Beydoun et al., 2012; Raj et al., 2017; Wirtz et al., 2018). Consistent abuse on the domestic front has physical health implications leading to long term trauma with cascading impacts on overall wellbeing, and health outcomes (Ackerson and Subramanian, 2008). Investigations into the relationship between natural disasters and IPV have not yielded consistent results. For instance, Frasier et al. (2004) did not find a significant increase in IPV after a disaster while Harville et al. (2011) and Anastario et al. (2009)

found some disaster experiences were associated with higher likelihood of conflict and violent means of resolution. Data on both disasters and IPV are scarce owing to the unpredictability of disasters and shame associated with reporting violence, resulting in largely under-reported numbers (Masson et al., 2016; Rezaeian, 2013; Weinstein et al., 2000). This paper contributes to this body of literature and examines the association between disasters and IPV by examining trends in southern India six years before, a year after, and a decade after the 2004 tsunami.

2. Literature review

The Indian Ocean Tsunami struck peninsular India in 2004 killing more than 15,000 and affecting thousands more (CRED, 2019; Shaw, 2015). TamilNadu and Andhra Pradesh on the east coast, and Kerala on the west bore the brunt of the disaster; Karnataka was not directly affected (Sheth et al., 2006). Despite different languages and intersecting cultures, the four states are often seen as a geographical unit within mainland India and fare higher on most social indicators compared to other states (Mukherjee et al., 2014; Oomen, 2009). Kerala outperforms

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on social indicators; post economic reforms, Tamil Nadu, Karnataka, and Andhra Pradesh rank higher on financial infrastructure than Kerala (Ghosh and De, 2004). Early assessment reports after the tsunami indicated economic stress, relocation, mental health impacts, and inequities in recovery experiences (Rodriguez et al., 2006). Relative distance from the tsunami as well as local topography played a critical role in buffering from the effects of the disaster (Rodriguez et al., 2006; Sheth et al., 2006). Disruptions in the aftermath of disaster lead to loss of social networks and higher risk of gendered psychological distress among women (David and Enarson, 2012). Studies have investigated associations between stress from natural disasters and increased IPV in the US after hurricane Floyd in 1999 and Hurricane Katrina in 2005 (Frasier et al., 2004; Harville et al., 2011). Weitzman and Behrman (2016) examined how disasters affect intra-family dynamics to identify theoretical pathways between disasters and IPV and found increased probability of physical and sexual IPV two years after disaster.

IPV encompasses physical, sexual, and emotional abuse, as well as controlling behaviors by an intimate partner. (WHO, 2012). Violence against women is a product of gendered power regimes and often a function of socio-economic contexts. A survey of literature reveals several individual and community level predictors of IPV. Socio-economic and demographic characteristics like age, income, religion, and education are known predictors of IPV and have been theoretically validated (Ackerson and Subramanian, 2008; Brahmapurkar, 2017; Kalokhe et al., 2017; Wirtz et al., 2018). In India, societal inequalities owing to patriarchy, caste, and class are closely linked to women's experience of IPV (Kalokhe et al., 2017; Krishnan, 2005; Mahapatro et al., 2012). Critical to note here is that historic marginalization of minority religions and castes in India has led to disparities on key indicators of education, income, and social status (Jahan, 2016). The resulting power differential produces disparate life-outcomes. Violence within these communities should be situated in context of structural power differences, where gender, caste, and religion intersect with poverty and inequitable access to economic and social resources (Krishnan, 2005; Sokoloff and Dupont, 2005).

Alcohol misuse is an empirically proven predictor of IPV with higher severity of violence observed when either partner had been drinking (Graham et al., 2011; Stanley, 2012; Wilson et al., 2014). Both alcohol and tobacco usage operate within the power differential in social structures and co-occur with violence and psychosocial stress (Ackerson et al., 2007; Graham et al., 2011). While rates of domestic violence are similar across rural, peri-urban and urban areas, violence is more chronic in rural, and poorer urban areas owing to higher rates of substance abuse, lower education, higher unemployment, and the overlapping power differences (Edwards, 2015; Krishnan et al., 2010).

3. Conceptual framework

Disaster vulnerability theory explains the susceptibility of certain individuals, groups, and communities to heightened losses in a natural disaster (David and Enarson, 2012; Zakour and Gillespie, 2012). For women, vulnerabilities include increased mortality, loss of economic and social ties, gender-based violence including IPV, and restricted access to reproductive, mental, and sexual health facilities (Enarson, 1998; Neumayer and Plümper, 2007; Nguyen, 2018). Within this context, the paper focuses on one gendered experience of disasters- IPV.

Vulnerability theory was advanced by O'Keefe et al. (1976) in their seminal paper where they suggested that not much was natural in the makings of a natural disaster because disasters were made up of social factors. A central theme within this framework is that disasters affect different groups differentially, suggesting a combination of biophysical and social processes, long-term and immediate, that make communities or individuals more susceptible to harm (Wisner, 2016). Zakour and Gillespie (2012, p. 10) posit that since disasters are not preventable by themselves, the way to minimize losses and accelerate recovery after a disaster is to decrease vulnerabilities of certain populations.

Pongponrat and Ishii (2018) examine multiple points of marginality among Thai women in Japan during the Japanese tsunami, such as historical and ethnic backgrounds, minority or immigration status, age, educational achievement, and dependence on their Japanese husbands. These factors pre-and-post-tsunami were critical to understand why Thai women were marginalized in the disaster aftermath. According to vulnerability theory, women's experience of violence should be viewed as a continuum to understand it in the context of humanitarian emergencies (Aolain, 2011; Durant et al., 2011; Enarson, 2012; Jayawardhan, 2017; Zakour and Gillespie, 2012).

The conceptual model guiding this study (Fig. 1) looks at socio-economic, demographic, and behavioral factors affecting women's vulnerability. The state, as proxy for experience of disaster is the exposure variable to examine associations between state of residence and contextual vulnerabilities on the outcome, IPV.

Vulnerability theory outlines the importance of systemic and societal factors in determining disaster outcomes (Zakour and Gillespie, 2012). First, First, and Houston (2017) emphasize the need to take various transects of socio-cultural inequalities into account in studies on IPV. This paper addresses this need and looks at determinants of IPV among women in India (age, education, income, religion, caste, place of dwelling) and superimposes state as proxy for experience of tsunami in 2004. Two main questions emerge:

1. How did prevalence of IPV change across southern states of India after the 2004 tsunami?
2. How do contextual vulnerabilities factor into risk of IPV among women?

Primary hypotheses for this study are:

Ha1. States impacted by the tsunami (Kerala, TamilNadu) will have heightened IPV in the immediate aftermath of disaster compared to states with low impact (AndhraPradesh) and no impact (Karnataka).

Ha2. Socio-economic and demographic vulnerabilities put women at higher risk of IPV in both disaster and non-disaster scenarios.

4. Methods

4.1. Data and sample

The principal data are from National Family Health Surveys (NFHS) 1999, 2006, and 2015 (IIPS, 2000, 2007, 2017). The NFHS are large-scale, multi-round surveys, part of the Demographic and Health Surveys (DHS), carried out by International Institute of Population Sciences (IIPS), India. Data are nationally representative and provide information on demographic and health indicators. For this study, individual data pertaining to women aged 15–49 were merged with household member data to include household level variables. Sample size was limited to respondents in four southern states: TamilNadu, Karnataka, AndhraPradesh, and Kerala. In 1999 (wave1), women who responded to the question of whether or not they experienced physical violence in the last 12 months were included in the analysis. For 2006 (wave2) and 2015 (wave3), ever-married women who answered the DHS domestic violence module were included. Wave1 data were collected between 1998 and 1999, wave2 data in December 2005–August 2006, and wave3 between 2015 to 2016.¹ Response rates were 96% for wave1, 95% for wave2, and 97% for wave3. Listwise deletion produced eligible cases of women residing in the four states of interest, who answered the domestic violence question or module.

¹ Tables 1.A in appendix details data collection time and month.

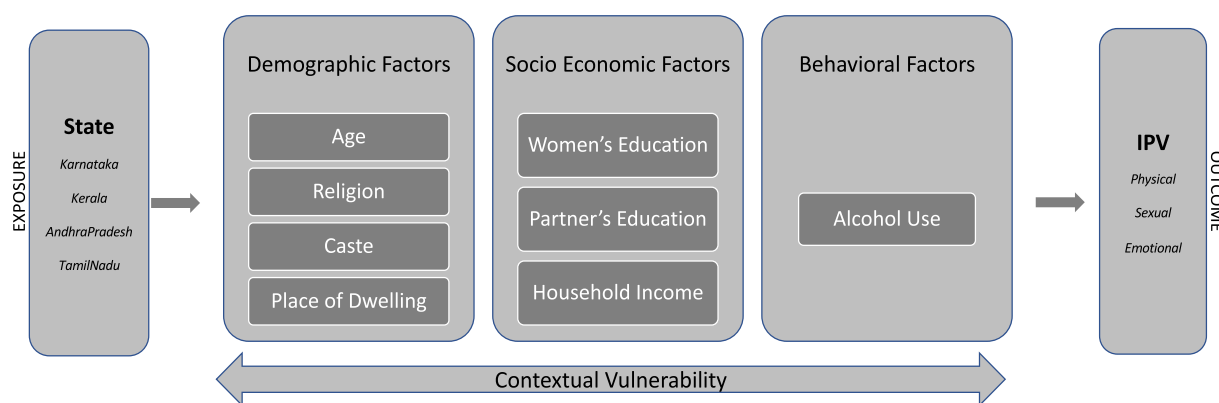


Fig. 1. Conceptual Model for effect of state of residence and contextual vulnerabilities on outcome variable (IPV).

4.2. Measures

Dependent variable. The dependent variable for this study, Intimate Partner Violence (IPV) is a dichotomous variable describing whether a woman experienced any form of IPV in the last 12 months or not. The dependent variable in wave1 measured if a woman experienced physical beatings in the last 12 months coded 1–4, “many times”, “few times”, “once”, and “not beaten”. This was dichotomized into 0 (no) and 1 (yes) to yield physical violence experienced in the last year. Beginning with wave2, NFHS administered a domestic violence module to a randomly selected subsample. Wave2 data were collected in 2006 and include self-reported IPV in the 12 months before survey, in the months immediately following the tsunami. In waves 2 and 3, IPV comprises physical, sexual, and emotional violence (WHO, 2012). Seven variables measured physical violence: being pushed, slapped, punched, kicked, strangled, twisted arm, and threatened with knife by spouse. Each variable was dichotomized; anyone who said yes was coded 1(Yes) and anyone who responded no, or not in last 12 months was coded 0(No). Two variables, forced sex and forced sexual acts were similarly dichotomized to form the sexual violence variable. Three variables, humiliated, threatened, and insulted by spouse; were dichotomized into emotional violence. These three kinds of violence were combined into IPV where respondents who answered “Yes” on any of the items were coded 1 and anyone who answered “No” was coded 0. Cronbach's Alpha for these twelve items was 0.82 in wave2 and 0.85 wave3, showcasing strong internal consistency.

Independent Variable. The Indian Ocean tsunami struck in the early hours of December 24, 2004 affecting major parts of the Indian subcontinent. TamilNadu and Kerala were impacted severely, AndhraPradesh moderately, and Karnataka was not directly affected. Women's state of residence is used as proxy for experience of disaster. The original state variable had 29–33 states across waves owing to newer states created. Southern states of Kerala, AndhraPradesh, Karnataka, and TamilNadu were retained for analysis in all three waves. Karnataka (coded 3) was used as the reference category for the statistical analyses.

A number of co-variables were included to examine association between state of residence and IPV informed by vulnerability theory and the literature reviewed.

Socio-Economic Factors. Women's and partner's education were categorical variables, with four categories: no formal-schooling, primary, secondary, and higher education; the first category being the reference group. Wealth index, a country specific measure of the household's cumulative living standard determined by the DHS, includes assets, water access, housing material etc. (NFHS3, 2010). This variable is measured in quintiles from poorest to richest with the poorest quintile used as the reference group.

Demographic Factors. Age, a continuous variable, measured current age of the respondent. Religion was recoded into three categories,

Hindus (coded 0), Muslims (1), Christians and Others (2) that included minority religions in India. Caste was recoded into three categories: Scheduled Caste/Scheduled Tribe (SC/ST) (0), Other backward castes (OBC) (1), and ‘others’ labeled General (2) which included higher castes and those that did not identify as SC/ST or OBC. Vart, Jaglan, and Shafique (2015) use NFHS2 (wave1) and note the category ‘others’ as least disadvantaged; this category is used as the reference group. Place of dwelling was dichotomous, coded Urban (0) and Rural (1).

Behavioral Factors. There are variations in reporting between surveys; wave1 asked alcohol related questions to any household respondent, and waves 2 and 3 asked them separately to women. In wave1, alcohol use of respondent, and in waves 2 and 3, partner's alcohol use was dichotomized to yes (1) and no (0); missing values were dropped.

4.3. Data analysis

Descriptive data from the three waves were reviewed to gauge prevalence of violence across states. Data from wave2 were visualized to note IPV prevalence in the year after tsunami across the southern states. To answer the first research question, state of residence was included as the primary exposure variable. For analytic samples across waves, three logistic regression models were run controlling for state of residence and other covariates. In all three waves, models controlling for state of residence had a lower Bayesian Information Criterion (BIC) showing strong support for the full model with the state as the main independent variable of interest (as proxy for experience of disaster). To answer the second question, 12 additional state-level logistic regression models were run across the three waves, controlling for socioeconomic and demographic factors. For state level analysis, behavioral covariates on alcohol use were not included. Domestic violence weights, provided by DHS to compensate for non-response rates and to ensure the subsample remained representative, were applied to analyses in waves 2 and 3. Sample weights were employed for wave1. Quantum Geographical Information Systems (QGIS3) was used to represent IPV in wave2; Stata version 15 was used for all data management and statistical analyses (StataCorp, 2017).

5. Results

Within analytic samples, 51% women and 38% of their partners in wave1, 31% women and 27% men in wave2, and 21% women and 18% men in wave3 had no formal schooling. More than 80% across waves were Hindus and 12–16% belonged to minority religions. 41.5% of the analytic sample in wave1 was in the lowest two wealth quintiles. About 25% in wave2 and 15% in wave3 belonged to the poorest quintiles. More than 80% of the samples across waves identified as either SC/ST or OBC. More than half lived in rural areas though percentage of rural population declined over time. Weighted descriptive statistics can be

Table 1
Weighted Descriptive Statistics for the three waves in the study.

Variable	Wave1 (1999)		Wave2 (2006)		Wave3 (2016)	
	(N = 3973)		(N = 12,912)		(N = 7997)	
	Frequency	%	Frequency	%	Frequency	%
Violence						
Yes	1772	45.68	2850	21.22	2432	31.31
No	2201	54.32	10,062	78.78	5565	68.69
Women's Education	3973					
No formal-schooling	2035	51.22	4064	36.95	1684	21.33
Primary	922	23.21	2185	16.92	1035	13.1
Secondary	861	21.67	5418	41.96	4168	51.27
Higher	155	3.90	1245	9.64	1110	14.3
Partner's Education	3973					
No formal-schooling	1413	38.3	3016	26.8	1437	18.17
Primary	993	25.09	2114	17.68	1118	13.79
Secondary	1226	28.54	6083	44.77	4293	53.12
Higher	341	8.06	1699	10.75	1149	14.92
Wealth						
Poorest	648	18.21	992	8.90	304	2.84
Poorer	892	23.37	1790	16.28	1247	12.97
Middle	1087	28.14	2954	26.41	2189	26.63
Richer	900	20.58	3651	26.30	2502	31.60
Richest	446	9.69	3525	22.10	1755	25.94
Religion						
Hindus	3414	86.65	10,600	84.14	6593	83.07
Muslims	343	7.83	1544	10.21	874	10.59
Christians & others	216	5.52	768	5.65	530	6.33
Caste						
SC/ST	1214	31.59	2840	22.67	2107	26.55
OBC	2137	53.05	7173	57.81	4950	61.4
General	622	15.36	2899	19.53	940	12.05
Place						
Urban	1236	25.13	4946	38.15	3186	44.54
Rural	2737	74.87	8018	61.85	4811	55.46
Alcohol						
No	3817	94.86	7535	58.65	5083	62.97
Yes	156	5.14	5377	41.35	2914	37.03
State						
Kerala	292	5.70	1733	11.42	1367	17.09
AndhraPradesh	928	32.97	4203	37.2	1053	13.17
Karnataka	928	18.15	3175	25.48	2037	25.47
TamilNadu	1825	43.17	3801	25.89	3540	44.27
Woman's Age	Mean	SD	Min	Max		
Wave1	32.24	8.60	15	49		
Wave2	32.47	8.49	15	49		
Wave3	34.28	8.42	16	49		

observed in Table 1. In wave1, women were asked questions on physical beatings in the last 12 months which could include non-intimate partner physical violence. 46% of the women had experienced physical violence in the past year in wave1. Subsequent surveys included information on other forms of IPV which could be a reason for the un-intuitive drop in physical violence prevalence in wave2. In wave2, 21.22% of the analytic sample had experienced IPV and the prevalence was 31.31% for wave3.

IPV increased between 2005 and 2015 in all states except Kerala. This increase, despite the recording system being similar in waves 2 and 3 could be from more cases being reported due to increased awareness, or more robust interviewing. TamilNadu and Karnataka had the highest prevalence of physical violence in the twelve months preceding wave1. In waves 2 and 3, highest IPV levels were in TamilNadu and AndhraPradesh. Interestingly, the prevalence of sexual and physical violence doubled in nearly all states except Kerala between waves 2 and 3. Emotional violence saw some more fluctuations comparatively between waves 2 and 3. Table 2 shows trends in IPV prevalence in the analytic sample across waves and the percentage change from wave2 to wave3.

Fig. 2 shows the percentage of women experiencing physical, sexual, or emotional violence was much higher in AndhraPradesh (26%), and TamilNadu (28%), both of which bore the brunt of the tsunami in 2004. Karnataka and Kerala show lower levels of self-reported violence (about 14%) despite being differentially affected by the tsunami (IIPS, 2007). Wave2 data are visualized in Fig. 2.

Table 3 shows results of logistic regression models for analytic samples across the three waves controlling for state of residence. In wave1, compared to Karnataka, women in AndhraPradesh were associated with 37% increased odds of physical violence in the past year, adjusting for other covariates in the study ($p < 0.01$). In wave2, respondents in all three states affected by the tsunami were associated with higher risk of IPV compared to those residing in Karnataka. Compared to Karnataka, women in TamilNadu had 98% higher odds of experiencing IPV ($p < 0.001$), holding other variables constant. In AndhraPradesh, moderately affected by the tsunami, the odds of a woman experiencing IPV were 85% higher ($p < 0.001$) and in Kerala, the odds were 41% higher ($p < 0.05$) when other variables were adjusted for. In the decade after the disaster, AndhraPradesh and TamilNadu were both associated with higher odds of IPV compared to

Table 2
Percentage of women experiencing violence in last 12 months in 4 southern states across survey years.

	1999					2005					2015					% change (2005–2015)			
	Phys.	Phys.	Sexual	Emo.	All	Phys.	Phys.	Sexual	Emo.	All	Phys.	Phys.	Sexual	Emo.	All	Phys.	Sexual	Emo.	All
Analytic Sample	45.68	18.68	2.55	8.45	21.22	26.28	5.27	15.42	31.31	40.69	106.67	82.49	47.55						
TamilNadu	40.34	20.53	2.01	10.41	27.96	33.09	6.67	19.05	39.33	61.18	231.84	83.00	40.67						
Kerala	34.81	9.66	3.76	6.9	14.21	8.14	2.86	6.2	11.65	–15.73	–23.94	–10.14	–18.02						
AndhraPradesh	55.32	24.48	2.54	9.57	26.1	34	4.25	16.98	37.14	38.89	67.32	77.43	42.30						
Karnataka	46.01	12.37	2.68	5.49	14.29	16.23	5.39	12.19	21.49	31.20	101.12	122.04	50.38						

Note: Wave1 measures physical violence; For waves 2 and 3 composite and different types of violence are shown.

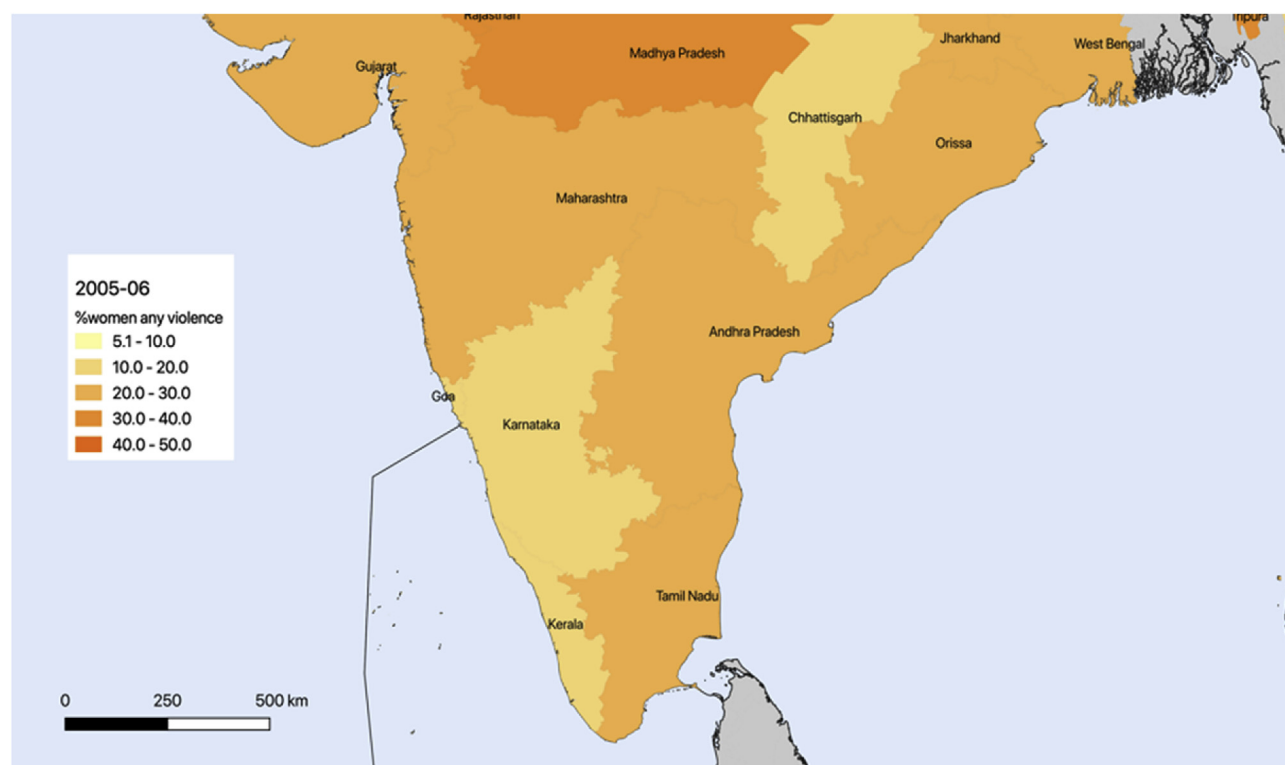


Fig. 2. Percentage of ever married women who experienced physical/sexual/emotional violence in 12 months before wave2 surveys.

Karnataka, adjusting for other covariates ($p < 0.001$). In the year after tsunami, an increase in woman's age was associated with marginally decreased odds of experiencing IPV with 95% CI between 0.97 and 0.98 ($p < 0.001$), adjusting for other variables in the study; ten years after the tsunami, age did not significantly predict odds of IPV. Women's education was not associated with violence in wave1, but in wave2, compared to those with no formal-schooling, women with higher than secondary education had 47% lower odds of experiencing IPV, controlling for other covariates ($p < 0.001$, 95% CI between 0.33 and 0.72). Partner's secondary education significantly reduced odds of IPV by 15% in wave2 ($p < 0.05$) controlling for other variables. In wave2, belonging to certain historically disadvantaged caste groups was associated with significantly higher odds of IPV compared to women belonging to the least disadvantaged groups ($p < 0.05$). In all three waves, an increase in wealth of a household was significantly associated with lower odds of IPV. In wave2, belonging to middle to richest wealth quintiles was associated with significantly lower odds of IPV compared to the poorest, adjusting for other variables in the study.

For the overall sample in wave2, compared to those practicing the majority religion, belonging to minority religions was associated with significantly higher odds of IPV, holding other variables constant ($p < 0.001$, $P < 0.05$). Religion was not a significant predictor for the analytic sample in wave3. Alcohol consumption-household member

response in the base year, and partner's consumption in subsequent waves was significantly predictive of increased odds of IPV, controlling for other covariates. In wave2, partner's alcohol consumption was associated with twice the odds of IPV compared to those whose partner did not consume alcohol ($p < 0.001$). Further, different types of violence were also regressed on the covariates across waves 2 and 3 and showed results consistent with the composite violence variable. Results of logistic regression analyses across survey years are presented in Table 3.

Per hypothesis 2, state level analyses were conducted to examine associations between socio-economic and demographic vulnerabilities (woman's age, education (women and partner), religion and caste, household wealth, place of residence) and IPV in each state.

1998–99. In wave1 in TamilNadu, compared to the majority religion, belonging to Islam was associated with 90% increased odds of experiencing IPV controlling for other variables ($p < 0.01$). Belonging to marginalized caste groups in TamilNadu was not associated with IPV. In Kerala and Karnataka, belonging to minority religions or caste was not significantly associated with IPV. In AndhraPradesh, belonging to certain minority religious groups was predictive of twice the odds of IPV compared to the reference group, controlling for other variables ($p < 0.05$). Place of residence was not significantly predictive of violence across states in wave1. In TamilNadu and Karnataka,

Table 3

Results of logistic regression predicting IPV by state and other covariates in Odds Ratio (Linearized SE in parenthesis).

Variables	1998–99	Intimate Partner Violence 2005–06	2015–16
Woman's Age	0.95*** (0.004)	0.98*** (0.004)	1.00 (0.005)
Muslim ⁺	1.40* (0.197)	1.57*** (0.175)	0.85 (0.144)
Christian and other ⁺	1.13 (0.188)	1.36* (0.197)	1.04 (0.149)
SC/ST ⁺⁺	1.07 (0.139)	1.32* (0.144)	1.24 (0.205)
OBC ⁺⁺	0.95 (0.114)	1.02 (0.096)	1.05 (0.143)
Women's Education ⁺⁺⁺			
Primary	0.84 (0.080)	1.04 (0.096)	0.96 (0.109)
Secondary	1.00 (0.114)	0.92 (0.084)	0.83 (0.094)
Higher	0.77 (0.179)	0.53*** (0.095)	0.48*** (0.084)
Partner's Education ⁺⁺⁺			
Primary	0.95 (0.092)	1.16 (0.095)	1.05 (0.112)
Secondary	1.05 (0.106)	0.85* (0.068)	0.85 (0.094)
Higher	0.87 (0.146)	1.05 (0.146)	0.91 (0.142)
Wealth [@]			
Poorer	1.01 (0.112)	0.99 (0.099)	0.96 (0.159)
Middle	0.87 (0.097)	0.70*** (0.074)	0.76 (0.130)
Richer	0.61*** (0.079)	0.59*** (0.075)	0.66* (0.123)
Richest	0.51*** (0.090)	0.35*** (0.055)	0.54** (0.123)
Alcohol	1.66** (0.314)	2.91*** (0.201)	2.57*** (0.209)
Rural	0.89 (0.080)	0.91 (0.084)	0.80* (0.081)
State			
Kerala	0.84 (0.133)	1.41* (0.187)	0.68 (0.161)
AndhraPradesh	1.37** (0.137)	1.85*** (0.211)	1.99*** (0.399)
TamilNadu	0.92 (0.086)	1.98*** (0.214)	2.42*** (0.506)
Constant	5.62*** (1.227)	0.27*** (0.059)	0.38** (0.121)
Observations	3973	12,912	7997

Note: +++ Reference: No formal-schooling; + Reference: Hindus; ++ Reference: General; Rural: Reference: Urban; @ Wealth reference: poorest; State Reference: Karnataka. *p < 0.05, **p < 0.01, ***p < 0.001.

belonging to the highest wealth quintile was associated with lower odds of IPV as compared to the poorest fifth of the sample, holding other variables constant ($p < 0.05$). In Karnataka, compared to women with no formal-schooling, those with primary education had slightly lower risk of IPV controlling for other variables ($p < 0.05$) and in TamilNadu, compared to those with no formal-schooling, partner's higher than secondary education was associated with significantly lower odds of IPV ($p < 0.05$).

2005–06. Increase in women's age significantly predicted lower odds of IPV in TamilNadu and AndhraPradesh and slightly higher odds in Kerala. In TamilNadu, controlling for other covariates, belonging to Christianity was significantly predictive of higher odds of IPV compared to Hinduism, and no association was found for caste groups. In Kerala, holding other covariates constant, belonging to Islam was associated with 87% higher odds of experiencing IPV compared to the reference group ($p < 0.01$). Compared to least disadvantaged caste categories, belonging to SC/ST groups in Kerala was associated with heightened

odds of experiencing IPV, holding other variables constant ($p < 0.05$). Moreover, in states with zero to moderate impact of the tsunami, belonging to certain historically disadvantaged caste groups was associated with higher odds of experiencing IPV compared to least disadvantaged groups, controlling for other variables. Place of dwelling (rural) was not significantly predictive of odds of experiencing IPV in any state that experienced the tsunami, but in Karnataka (no impact), women in rural areas had 34% lower odds of experiencing IPV controlling for other variables. In TamilNadu, compared to those in the lowest wealth quintile, belonging to the top wealth quintiles was associated with significantly lower odds of IPV controlling for other covariates ($p < 0.01$, $p < 0.001$). Compared to women with no formal-schooling, higher than secondary education lowered odds of IPV in all states except Kerala.

2015–2016. Ten years after, religion and caste did not predict IPV in any of the four states. Compared to women in urban areas, those in rural areas were associated with 46% lower odds of experiencing IPV in Kerala and 41% lower odds in AndhraPradesh when controlling for other variables in the study, but urban/rural residence was not significantly predictive of IPV in TamilNadu or Karnataka. Women with higher education compared to those with no formal-schooling were associated with lower risk of IPV in all states except AndhraPradesh. The high coefficients and standard errors for women's education in Kerala need to be interpreted with caution. Kerala has the highest literacy in the country and the reference group of no formal education was the smallest, resulting in amplified odds ratios and standard errors. In TamilNadu, compared to the poorest, women in the middle and richest quintiles were associated with lower odds of IPV when other covariates were controlled.

In summary, compared to Karnataka, residence in states affected by tsunami was significantly associated with higher odds of IPV in the immediate aftermath of disaster, confirming hypothesis one. For the second hypothesis, IPV was regressed on multiple points of vulnerability. Results show that existing vulnerabilities put women at higher risk in states that experienced the disaster, as well as the state that did not, with some of these factors diminishing over time. These results and their import are discussed in the subsequent section.

Tables 4–6 show state-level logistic regression analysis output predicting IPV controlling for socio-economic and demographic factors for the three waves.

5.1. Model diagnostics and goodness of fit

The Hosmer-Lemeshow test for logistic regression was not statistically significant indicating good model fit across the waves (Wave1: $p = 0.48$; Wave2: $p = 0.65$; Wave3: $p = 0.07$). The linktest for model specification confirmed that the models across waves had no obvious specification errors and no irrelevant variables were included in the equation. Additional analysis found that 62% of the cases were correctly classified in wave1, 78.03% in wave2, and 72.36% in wave3, indicating moderately correct classification. The McFadden's R-squared for model 1 was 0.05, for model 2, 0.10 model 3 was 0.12. The area under ROC curve confirmed the models' modest predictive power across waves (wave1:0.66, wave2:0.72, wave3:0.73 under the ROC curve). There was no evidence of multicollinearity among the predictors across waves (Mean vif for wave1 = 1.78; wave2 = 2.15; wave3 = 2.83).

6. Limitations

Some limitations should be noted:

1. The survey data are cross-sectional; it is not possible to establish causality, rather, it helps identify trends in risk and prevalence of IPV through the three time-points.
2. Populations at each wave may differ especially if they were displaced in high impact states after tsunami. Patterns of internal

Table 4
Predicting IPV by state for wave1, 1999. Results in Odds Ratio (Linearized SE in parenthesis).

Variables	TamilNadu	Kerala	Karnataka	AndhraPradesh
Woman's Age	0.95*** (0.006)	0.97 (0.018)	0.95*** (0.008)	0.95*** (0.008)
Religion ⁺				
Muslim	1.90** (0.474)	1.47 (0.523)	0.79 (0.221)	2.08* (0.699)
Christian & other	1.37 (0.395)	1.35 (0.565)	0.64 (0.374)	1.09 (0.292)
Caste ⁺⁺				
SC/ST	1.20 (0.887)	1.50 (0.673)	1.28 (0.304)	0.92 (0.204)
OBC	1.03 (0.759)	0.80 (0.259)	0.90 (0.205)	0.89 (0.181)
Women's Ed. ⁺⁺⁺				
Primary	0.94 (0.125)	1.40 (0.749)	0.63* (0.132)	0.70 (0.143)
Secondary	1.19 (0.196)	2.23 (1.247)	0.89 (0.211)	0.63 (0.160)
Higher	0.70 (0.236)	2.47 (1.902)	0.65 (0.370)	0.83 (0.473)
Partner's Ed. ⁺⁺⁺				
Primary	0.91 (0.132)	1.25 (0.581)	0.96 (0.189)	0.87 (0.165)
Secondary	0.99 (0.148)	0.82 (0.376)	1.24 (0.243)	1.08 (0.227)
Higher	0.61* (0.147)	1.26 (1.009)	0.99 (0.366)	1.27 (0.448)
Wealth Quintile [@]				
Poorer	1.24 (0.225)	1.40 (1.166)	1.32 (0.269)	0.70 (0.137)
Middle	1.15 (0.204)	0.79 (0.639)	0.83 (0.181)	0.71 (0.142)
Richer	0.86 (0.174)	0.39 (0.336)	0.65 (0.169)	0.41*** (0.105)
Richest	0.58* (0.150)	0.23 (0.235)	0.46* (0.168)	0.50 (0.190)
Rural	0.97 (0.120)	0.60 (0.188)	0.80 (0.145)	0.99 (0.208)
Constant	3.48 (2.787)	1.59 (1.831)	5.64*** (2.268)	10.91*** (4.463)
Total	1828	292	929	928

Note: +++Reference: No formal-schooling; + Reference: Hindus; ++ Reference: General; Rural: Reference: Urban; @ Wealth reference: poorest; State Reference: Karnataka. *p < 0.05, **p < 0.01, ***p < 0.001.

Table 5
Predicting IPV by state for wave2, 2006. Results in Odds Ratio (Linearized SE in parenthesis).

Variables	TamilNadu	Kerala	Karnataka	AndhraPradesh
Woman's Age	0.98*** (0.007)	1.03* (0.014)	0.99 (0.007)	0.98** (0.006)
Religion ⁺				
Muslim	0.82 (0.310)	1.87** (0.432)	0.96 (0.228)	1.58** (0.267)
Christian & other	1.82** (0.402)	1.09 (0.281)	1.09 (0.338)	1.24 (0.310)
Caste ⁺⁺				
SC/ST	1.13 (0.679)	1.82* (0.422)	1.70* (0.371)	1.57** (0.257)
OBC	0.76 (0.455)	0.90 (0.158)	1.11 (0.232)	1.22 (0.159)
Women's Ed. ⁺⁺⁺				
Primary	1.21 (0.188)	1.11 (0.401)	0.90 (0.146)	0.86 (0.141)
Secondary	0.87 (0.130)	1.52 (0.514)	0.85 (0.166)	0.86 (0.134)
Higher	0.46* (0.138)	1.55 (0.709)	0.13** (0.084)	0.50* (0.166)
Partner's Ed. ⁺⁺⁺				
Primary	1.23 (0.189)	0.85 (0.298)	1.00 (0.174)	1.15 (0.137)
Secondary	0.80 (0.123)	0.65 (0.258)	0.68* (0.114)	0.91 (0.110)
Higher	0.71 (0.179)	0.73 (0.351)	0.71 (0.196)	1.07 (0.234)
Wealth Quintile [@]				
Poorer	0.81 (0.138)	0.70 (0.403)	1.03 (0.171)	1.08 (0.178)
Middle	0.58** (0.095)	0.53 (0.305)	0.67* (0.126)	0.79 (0.132)
Richer	0.54** (0.102)	0.38 (0.215)	0.55** (0.117)	0.63* (0.137)
Richest	0.36*** (0.094)	0.17** (0.098)	0.27*** (0.105)	0.36*** (0.091)
Rural	0.86 (0.132)	0.88 (0.172)	0.66* (0.129)	0.86 (0.142)
Constant	1.61 (1.122)	0.16* (0.139)	0.41* (0.175)	0.79 (0.232)
Total	3801	1734	3180	4206

Note: +++Reference: No formal-schooling; + Reference: Hindus; ++ Reference: General; Rural: Reference: Urban; @ Wealth reference: poorest; State Reference: Karnataka. *p < 0.05, **p < 0.01, ***p < 0.001.

- migration both inter and intra-state, follow many pathways and include a variety of factors such as livelihood, education, and marriage (Parida and Madheswaran, 2010). Post tsunami assessments suggest that much displacement was local but this has not been documented over time (Human Rights Center & East-West Center, 2005); and migration or displacement was not measured by NFHS.
- Waves 2–3 collected information on multiple types of violence, wave1 measured physical violence alone. This limits a comparison across waves. Wave1 provides important baseline information to assess trends over time; waves 2 and 3 can be directly compared.
 - The sample of women interviewed in wave1 included ever married women aged 15–49. In waves 2 and 3, while all women aged 15–49 were included, a subsample of ever married women was asked IPV questions on physical, sexual, and emotional violence from their partners.
 - IPV is difficult to measure owing to reliance on self-reporting by those experiencing it (Ackerson and Subramanian, 2008). Gage (2005) notes that self-reported data is susceptible to bias and is often difficult to verify; for a sensitive topic such as IPV, women might not be forthcoming due to shame, guilt, or fear. The NFHS followed the WHO's guidelines on ethical collection of domestic violence information and at the state level, women who reported domestic violence were provided with resources where they could get help

(IIPS, 2017).

- The McFadden's R-squared for models suggests low predictive power; the pseudo R-squares were primarily used to compare models with the state variable and without, to show the significance of state as an exposure variable.
- Analyses across waves were at the state level; NFHS do not include complete district level data. Since the tsunami mostly affected coastal areas, there will be differences by district owing to distance from the coast. Future research should examine district level differences to bolster data visualization and statistical inferences.
- Kerala, a high performing state on a number of development indicators, is an exception but was included to identify potential protective factors; moreover, even with lower levels of violence it is critical to see who might be at higher risk therein.

7. Discussion

The Indian Ocean tsunami wrought destruction on peninsular India with 15,000 mortalities and about 654,512 people affected (CRED, 2019). TamilNadu and AndhraPradesh on the east coast were directly impacted. On the west coast, Kerala saw extensive damage due to waves off the coast of Sri Lanka. The geological variations within these states and relative distance from exposure to the tsunami waves were a factor

Table 6
Predicting IPV by State for wave3, 2016. Results in Odds Ratio (Linearized SE in parenthesis).

Variables	TamilNadu	Kerala	Karnataka	AndhraPradesh
Woman's Age	1.00 (0.007)	1.05** (0.014)	1.00 (0.010)	0.99 (0.011)
Religion ⁺				
Muslim	0.82 (0.264)	1.08 (0.309)	0.57 (0.218)	0.66 (0.176)
Christian and other	0.98 (0.178)	1.35 (0.401)	1.64 (0.828)	1.01 (0.404)
Caste ⁺⁺				
SC/ST	1.57 (0.511)	1.76 (0.660)	1.16 (0.418)	1.05 (0.295)
OBC	1.15 (0.366)	0.86 (0.247)	0.92 (0.213)	1.00 (0.223)
Women's Ed. ⁺⁺⁺				
Primary	0.89 (0.140)	6.61* (6.121)	1.00 (0.202)	1.02 (0.211)
Secondary	0.85 (0.123)	6.81* (6.071)	0.61* (0.119)	0.70 (0.179)
Higher	0.40*** (0.084)	6.08 (5.598)	0.50 (0.212)	0.54 (0.254)
Partner Ed. ⁺⁺⁺				
Primary	1.03 (0.162)	1.09 (0.961)	1.04 (0.219)	1.09 (0.213)
Secondary	0.88 (0.138)	0.85 (0.766)	0.80 (0.152)	0.71 (0.163)
Higher	0.86 (0.181)	0.40 (0.418)	1.22 (0.391)	0.56 (0.199)
Wealth Quintile [@]				
Poorer	0.89 (0.213)	2.67 (3.069)	0.93 (0.256)	0.90 (0.348)
Middle	0.62* (0.140)	1.07 (1.230)	0.80 (0.239)	0.79 (0.322)
Richer	0.63 (0.154)	0.59 (0.674)	0.52 (0.191)	0.54 (0.225)
Richest	0.52* (0.136)	0.24 (0.287)	0.60 (0.359)	0.43 (0.189)
Rural	0.95 (0.109)	0.54* (0.129)	0.56 (0.193)	0.59* (0.126)
Constant	1.24 (0.637)	0.01** (0.017)	0.89 (0.502)	2.91 (2.008)
Total	3540	1367	2037	1053

Note: +++ Reference: No formal-schooling; + Reference: Hindus; ++ Reference: General; Rural: Reference: Urban; @ Wealth reference: poorest; State Reference: Karnataka. *p < 0.05, **p < 0.01, ***p < 0.001.

in how they were impacted (Sheth et al., 2006). The four southern states have a rich, interwoven cultural history and it makes empirical and theoretical sense to look at them as a geographical entity (Oomen, 2009). This paper investigates gendered impacts of the tsunami on women by examining prevalence and covariates of IPV before and after the disaster. Two hypotheses guided by vulnerability theory were tested. State of residence was included as a determinant of IPV to assess if states impacted by the tsunami had higher rates of violence compared to states with low or no impact. State of residence was significantly predictive of higher risk of IPV in all states that experienced the tsunami in the immediate aftermath. Overall IPV increased from 2005 to 2015 by about 48%. In wave2, the highest rates of violence were in TamilNadu and AndhraPradesh. Despite being impacted, Kerala had a similar rate of violence as Karnataka, which was not impacted (14%). Results are consistent with previous scholarship documenting increased violence against women post disaster, and lifetime increases in the years after (Anastario et al., 2009). These results are however, at odds with Amaral, Bhalotra, and Prakash's (2019) findings where no difference was found in reported IPV between 2005 and 2015 in urban India, even as reports of other crimes against women went up. Increased IPV levels in southern states a decade after the disaster merits attention and points to systemic issues such as women's economic insecurity and continued gender inequality (Parkinson, 2011).

Kerala was the only state where IPV levels dropped from 14% to about 12% and reported the lowest IPV levels across waves. Kerala remains understudied as a state in terms of IPV; research finds high rates of violence despite education and employment gains for women (Jin et al., 2014). Increased violence against women in the aftermath of disasters is well documented (Enarson, 1999; First et al., 2017; Masson et al., 2016; Weitzman and Behrman, 2016) and this paper adds a long term trend analysis to show that rates of violence were indeed high in the months after the tsunami in impacted states, and remained high even after a decade.

In-group differences were assessed at state-level to test the second hypothesis across waves. Results indicate that in the year after tsunami women from minority religions and historically disadvantaged caste groups were at higher risk of IPV in states affected by the tsunami but also in Karnataka which was not directly affected. The confluence of religion, caste, and place is peculiar to South-Asia and can be a complexifier in already fraught gender relations; a disruptive event like a natural disaster can be an added stressor (Kruks-Wisner, 2011; Mahapatro et al., 2012). Studies across contexts indicate that women belonging to social and cultural minorities encounter obstacles to education, health services, and resources due to structural and social inequities coupled with institutional barriers (First et al., 2017). Low levels of integration in society, high levels of chronic stress, and beliefs in male dominance are other psychosocial correlates that have been linked to domestic violence, and are relevant in understanding the link between IPV and vulnerabilities associated with membership to minority groups (Ellison et al., 1999). In times of disaster, existing inequalities as well as disparities in aid and relief responses often lead to differential outcomes, and this was noted in the case of the Indian Ocean tsunami as well (Aldrich, 2010; Hines, 2007). Consistent with previous scholarship, alcohol use was predictive of higher risk of IPV across waves, pointing to alcohol misuse interacting with stress, poverty, and social disadvantage (Ackerson and Subramanian, 2008; Neufeld et al., 2005; Stanley, 2012). While trends in the year after tsunami point to heightened vulnerabilities of certain minorities in the analytic sample, in-state analyses show that even in states with low or no impact, belonging to minority groups and being in poverty were associated with higher risk of IPV. Economic insecurity compounded by existing forms of social discrimination is a key determinant of post disaster vulnerability (Enarson and Phillips, 2008).

A decade after, caste and religious identity had no statistical associations with IPV in all four states, while socioeconomic factors continued to matter. A possible explanation could be social and economic improvements within subgroups in the subsequent decade. For instance, a qualitative study in TamilNadu nine months after the tsunami revealed that communities were more cohesive than before and showed a reduction in caste-based differences, as a result of relocation to shelters and settlements where formerly separated communities came to live together and resolve long-standing differences (Rajkumar et al., 2008). Further, better systems of support both formal and informal, such as those encouraging women to report crimes or seek redressal owing to heightened awareness and supportive policy frameworks as suggested by Amaral et al. (2019) could be another buffering factor, however, that does not explain the overall increase in IPV.

Ackerson and Subramanian (2008) note a socioeconomic gradient to IPV in their study with women in higher socioeconomic categories being less likely to report violence, further stating that disadvantaged castes continue to have lower socio-economic status despite affirmative policies. As a contrast to other disadvantaged castes, belonging to OBC was not significantly predictive of IPV in any model in the current study. This could ostensibly be due to upward economic mobility in this group, with younger cohorts doing better on a number of socio-economic indicators than before (Deshpande and Ramachandran, 2014; Jaffrelot, 2015). While belonging to marginalized caste groups continues to be a discriminating factor, the confluence of caste and class plays a critical role in increased social vulnerability (Ray-Bennett,

2009). Since caste and religion-based discrimination are far from over in India, future studies should look at combined effects of caste, class, and religion to discern the interacting effects of improved standards of living with factors of social vulnerability to document their association with IPV. Empirical investigations such as (Deshpande and Ramachandran, 2014) analysis on changes in standard of living among recent cohorts of OBC for other marginalized caste groups can help illuminate these transitions.

O'Keefe et al. (1976) prioritized precautionary planning over post disaster relief to help vulnerable populations resist disasters in the first place; this paper confirms the prudence of this approach. Improving women's access to education and economic independence that comes with it is critical to reducing gender based violence (Ackerson and Subramanian, 2008). Findings from this study reveal that contextual vulnerabilities increase the risk of IPV among women and this was ostensibly amplified in the aftermath of a major disaster. A gendered optic of natural disasters is key to understanding social and institutional contexts of women. For helping professions in community engaged work, this means creating systems of support for women especially those belonging to historically marginalized communities. For instance, access to psychosocial support services can mediate stress caused by existing gendered vulnerabilities compounded by a disaster (McGuire et al., 2018) and needs to be made available to socially disadvantaged groups. Since disasters are difficult to predict, building these systems in non-disaster situations is crucial for their availability in times of added stress.

Using publicly available survey data to analyze trends in the absence of disaster data is helpful in bridging methodological and ethical

bottlenecks. Current data limits a causal analysis and does not include all possible explanatory variables such as changes in employment and residential status post disaster that could add to risk of IPV (Krishnan et al., 2010). Robust longitudinal data-collection at the local administrative level can aid future disaster and gender scholarship. Results of this study act as a baseline for South India for future state-level analyses. The overall increase in IPV over time merits attention of policy makers. Social welfare policies and disaster response strategies need to address enduring inequities to succeed in redressal, reconstruction, and positive societal change (Anastario et al., 2009). This paper contends that while IPV should be viewed in a continuum, a major natural disaster such as the Indian Ocean tsunami of 2004 could have exacerbated IPV for some women owing to long-standing vulnerabilities.

CRedit authorship contribution statement

Smitha Rao: Conceptualization, Formal analysis, Methodology, Software, Visualization, Writing - original draft, Writing - review & editing.

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Appendix

Table 1
Field Work (Source: National Family Health Surveys).

State	Wave1		Wave2		Wave3	
	From	To	From	To	From	To
AndhraPradesh	11/98	3/99	12/05	4/06	5/15	8/15
Karnataka	3/99	9/99	4/06	7/06	2/15	7/15
Kerala	3/99	7/99	4/06	8/06	3/16	10/16
TamilNadu	3/99	6/99	4/06	7/06	2/15	6/15

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